UNITED NATIONS ENVIRONMENT PROGRAMME

UNEP

Model United Nations Conference - Global

FROM THE COMMITTEE BUREAU

The United Nations Environment Programme bureau welcomes delegates to MUNCGLOBAL's flagship conference in Ghana, MUNC-GH.

This Background Guide presents the dual challenge of cross-border e-waste flows and the urgent need for greener, more resilient cities, offering background, legal frameworks and policy options for circular economy and urban design. The UNEP committee is convened to address two urgent environmental challenges: the surge in electronic waste (e-waste) and the need for greener cities. Discarded electronics contain toxic substances that pollute land and water if not managed properly, and rapid urban growth strains natural resources and infrastructure. Delegates should examine policies for responsible production, recycling and cross-border controls on e-waste.

Simultaneously, they must explore how cities can grow in harmony with nature – for instance, through sustainable building, green spaces and waste reduction. Study the guide to prepareactionable proposals on waste regulation, producer responsibility and sustainable urban policies that protect ecosystems and human health. Contact the bureau for any clarifications. We look forward to creative, implementable solutions.

TOPIC BACKGROUND

Electronic Waste: Modern electronics (phones, computers, appliances) become obsolete very quickly. Globally, people generate about **62 million metric tons of e-waste per year** – enough to fill 1.5 million garbage trucks annually. This is one of the fastest-growing waste streams. E-waste contains valuable materials (metals, plastics) but also hazardous substances (lead, mercury, flame retardants). If dumped improperly, these toxins can **pollute soil, groundwater and air.** The environmental and health harm is severe: WHO notes that e-waste pollution can cause developmental delays in children and other long-term health problems.

Currently only about **20% of e-waste is recycled properly**. The rest often ends up in landfills or informal recycling (by unprotected workers, sometimes in developing countries). The economic costs of poor e-waste management are high – an estimated **$78 billion per year in hidden health and environmental damage.** However, properly collecting and recycling e-waste can also yield benefits. A recent report found that investing in e-waste collection and recycling infrastructure could generate some **$38 billion in annual economic benefits by 2030** (from recovered materials, jobs, and avoided damage).

International trade in e-waste is a key issue. For decades, rich countries have illegally exported e-waste to poorer countries. In 2022, high-income nations shipped **3.3 billion kg of used electronics** to middle- and low-income countries, often without proper controls. To combat this, the Basel Convention (the UN treaty on hazardous waste) recently added a new regulation: as of January 2025, any export of e-waste now requires prior informed consent of the receiving country.

This amendment is a "landmark step" according to UNEP, but its success depends on enforcement. Delegates should consider how to implement such rules and help developing nations build e-waste facilities.

Possible solutions include: international standards for product design (making electronics more durable and repairable); producer responsibility laws (requiring manufacturers to take back old products); global e-waste tracking; and funding for recycling programs. These "upstream" strategies are widely seen as more effective than simply discarding waste. Encouraging a circular economy – where materials are reused – can reduce waste and create green jobs

Sustainable Urban Planning: Today over half the world's population lives in cities, with forecasts of continued urban growth. Cities concentrate demand for buildings, transport, energy and resources. As UNEP notes, buildings alone account for **one-third of global greenhouse gas emissions and one-third of waste.** Without sustainable planning, expanding cities can exacerbate climate change, pollution, and habitat loss.

Sustainable urban planning involves designing cities that minimize environmental harm. Key elements include: **green infrastructure** (parks, urban forests, green roofs) to reduce heat and flood risks; **efficient buildings and transit** (promoting energy-efficient housing, public transport, renewable energy); **waste reduction and recycling;** and **inclusive zoning** (avoiding slums, ensuring access to services). For example, natural spaces in cities can act as carbon sinks and cooling zones.

UNEP highlights that nature-based solutions (like urban trees and wetlands) can cut emissions, mitigate floods, and improve residents' health

Another focus is reducing resource use. Cities consume vast materials: one report found that by 2050 cities could **halve their resource consumption** through measures like building with less material, using renewable energy, and recycling construction waste. Waste management is also critical: municipal waste and plastics need systems of "reduce, reuse, and recycle." UNEP's own campus in Nairobi exemplifies this by building on already-used land and recycling materials.

Delegates should build on these ideas: for instance, regulations could require energy-efficient building codes, support sustainable transport, or finance green urban projects. Internationally, sharing best practices (e.g. "smart city" technologies or community-based recycling) can help cities globally.Because this topic connects to Agenda 2030, delegates may link e-waste and urban planning to several Sustainable Development Goals (e.g. SDG 11 on sustainable cities, SDG 12 on responsible consumption, SDG 13 on climate). UNEP encourages integrative solutions: for example, better waste management in cities reduces pollution and carbon emissions. The committee should seek policy coherence considering that electronics recycling itself is an urban issue and promote both local action (city-level planning) and international support (technology transfer, funding for green cities in developing countries).

COMMITTEE BACKGROUND

The United Nations Environment Programme (UNEP) is the UN's authority on environmental issues. Established in 1972, UNEP's mission is to coordinate global environmental activities and help governments implement sound policies. Today UNEP is headquartered in Nairobi and includes the **UN Environment Assembly (UNEA)** – the world's top environmental forum. The UNEA meets every two years with **universal membership of all 193 UN Member States.** It makes key decisions and recommendations on the environment.

UNEP's role includes scientific assessments, policy guidance, and convening stakeholders (governments, NGOs, businesses). It has led international agreements (e.g. the Montreal Protocol on ozone) and coordinates environmental data (e.g. the Global E-waste Monitor). UNEP works closely with others: the Basel, Rotterdam and Stockholm conventions (which govern hazardous waste) are now administered under UNEP leadership. For e-waste, the Basel Convention – attended by 190 countries – is especially relevant.

In practical terms, this committee simulates a UNEA or related environmental plenary. The bureau (chairs) will guide you as if you were environment ministers. Your "delegations" represent Member States. WHO and FAO delegates won't be here, so discussions focus on environmental regulation and urban development. However, you may mention other agencies when proposing partnerships (e.g., working with UN-Habitat on cities).Recent UNEA sessions have dealt with plastic waste, chemicals and climate. For e-waste, UNEP has published a 2020 E-waste Monitor report and led the Basel amendment. For urban planning, UNEP often collaborates with UN-Habitat (the UN's urban agency).

Note that sustainable cities also involve climate adaptation (e.g. including cities in national climate plans was highlighted by UNEP). Delegates should be aware of UNEP's emphasis on the "triple planetary crisis" (climate, biodiversity, pollution) and ensure proposals do not solve one problem while worsening another (e.g. e-waste recycling must not create new pollutants).

When drafting resolutions, consider UNEP's typical actions: it can sponsor technical guidelines, set voluntary targets, or launch capacity-building programs. Binding obligations (like trade bans) come from separate treaties (e.g., Basel). Nonetheless, a UNEA resolution can call for all countries to ratify the Basel Convention amendment on e-waste, or fund a global recycling initiative. For cities, UNEP might develop model policies or fund pilot projects. Delegates should aim for clear commitments, such as setting e-waste reduction targets or requiring environmental impact assessments for new urban projects.

COMMITTE STRUCTURE

The UNEA (or UNEP Council in older terms) includes all UN Members (193 states). The committee will be chaired by an elected President and officers, as per UN rules. Debate follows UNGA-style procedure. You may form working groups to combine ideas on e-waste and urban planning, since the issues overlap (e.g. urban recycling systems). Positions often align by development level: developed countries may emphasize advanced recycling technology and corporate responsibility, while developing countries stress financial and technical support.

The timeline is to research national laws on e-waste and urban planning. Many countries have e-waste laws or extended producer responsibility schemes; cite your country's example. Regarding cities, note local government roles (some countries require city planning codes, others don't).

Draft resolutions should balance environmental protection with economic feasibility. Commit to mechanisms like reporting on e-waste flows or setting up an international fund to upgrade recycling infrastructure. Remember to use the formal UNEA language (e.g. preambulatory and operative clauses) and to refer to UNEP publications and relevant treaties. Collaboration between countries, industry and civil society will be key themes.